Hexapod (Dancing Robot)

Team 6

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1 Project Description

Our main design is a rectangular shaped hexapod, featuring 6 legs with 3 degrees of freedom (DOF) each, allowing for the most optimized movement. The hexapod is designed to allow up to 2 riders at one time. People to sit and perform on both sides, with the emergency stop at the dead center of the robot, allowing for quick and easy access for both people. We have also planned safety precautions and emergency shut off, in case the robot flips.

The robot will be controlled via a custom wireless controller with joysticks and a speed dial, communicating over Bluetooth to an onboard Raspberry Pi. The design prioritizes safety in a dance performance environment, modularity for transport, and robust construction capable of supporting a 170 lb rider at 2–3 ft off the ground.

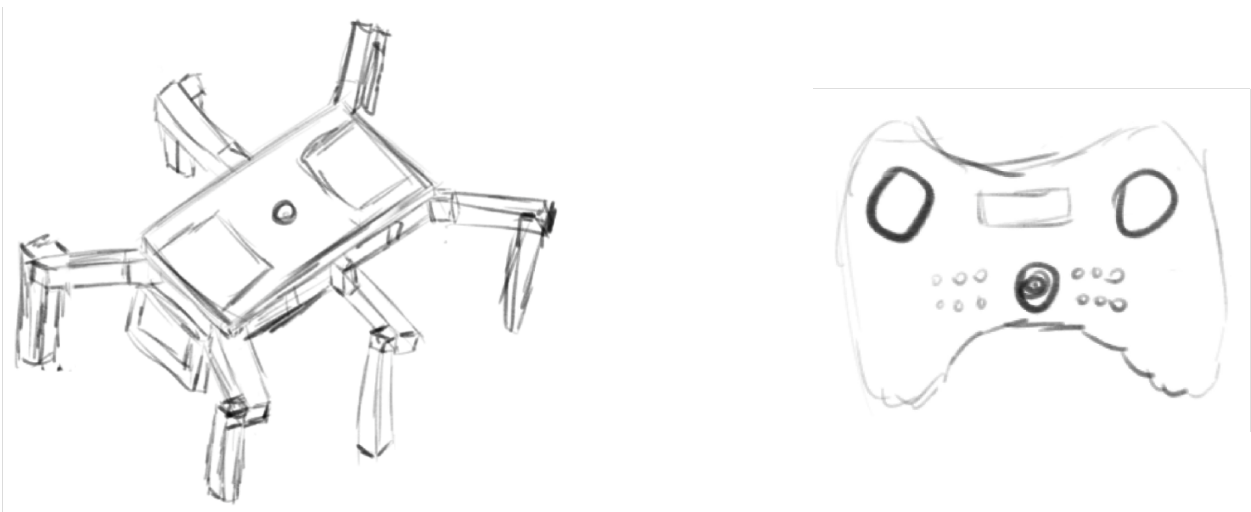

Figure 1: Concept sketches of the hexapod dancing robot (left) and custom wireless controller (right).

2 Design Requirements

2.1 Locomotion and Maneuvering

- Must be controlled/initiated through legs.
- Must travel in at least 2 directions (forward and backward).
- Must have zero-degree turning radius (turning in place).
- Must navigate to a designated point within 1 minute.

2.2 Size, Weight, Payload, and Rider Geometry

- Footprint must be less than or equal to 5 ft $\times$ 5 ft.
- User rides 2–3 ft off the ground.

- The device must be less than 150 lbs.
- Must carry a person up to 170 lbs.
- Seat must allow users to sit cross-legged or standard chair seated.

2.3 Packaging and Transport

- Must ship in a 3 ft $\times$ 3 ft case.
- Must disassemble into no more than 7 parts.

2.4 Power

- Battery life must be at least 30 minutes.

2.5 Safety

- Must be safe in a dance performance environment, no rough edges/burrs/exposed screw tips.
- Ride must be smooth enough that the user can ride safely without being strapped in.
- Emergency stop button accessible to both riders, positioned dead center.
- Auto shut-off if the robot flips (IMU-based detection).

3 Functional Architecture

The functional architecture of the hexapod dancing robot is organized into five major subsystems: Power, Compute, Mechanical System, Control, and Locomotion. Figure 2 depicts the block diagram showing the flow of energy, information, and mechanical forces between these subsystems.

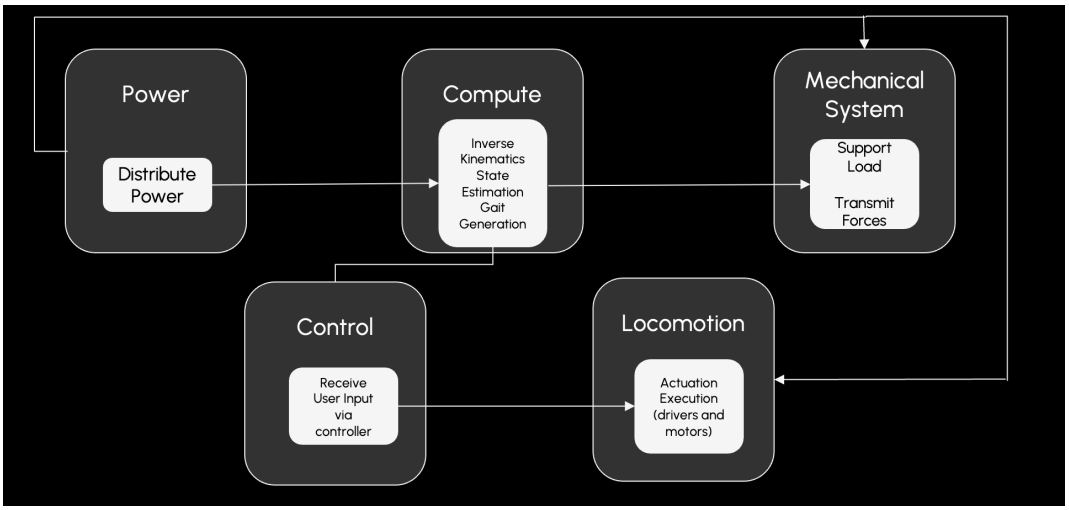

Figure 2: Functional Architecture Diagram of the Hexapod Dancing Robot.

Power distributes electrical energy from the onboard battery to all electronic and mechanical subsystems.

Compute performs inverse kinematics calculations to determine joint angles for each leg, estimates the robot's state (orientation, balance) using IMU data, and generates gait patterns for walking and dancing movements.

Control receives user input from the wireless controller (joystick directions, speed via potentiometer, and button commands) and translates these into high-level movement commands passed to the Compute subsystem.

Locomotion executes the computed joint angles by driving the 18 servo motors (3 per leg $\times$ 6 legs) through motor drivers, converting electrical signals into physical leg motion via actuation execution (drivers and motors).

Mechanical System provides structural support for the payload (riders), supports the load, and transmits the forces generated by the servo motors through the leg linkages to the ground.

4 Design Trade Studies

We have debated between multiple different designs for each part of the robot and the tradeoffs within. For each subsystem, we use a weighted decision matrix to evaluate and choose between design alternatives.

4.1 Leg Configuration: 2 DOF vs. 3 DOF

The number of degrees of freedom per leg is a critical design choice affecting mobility, complexity, and cost. A 2 DOF leg provides only a lift and swing motion, while a 3 DOF leg allows for more maneuverability leading to smoother and faster lateral movement and in-place turning.

Figure 3: Design alternatives for leg DOF: 2 DOF (top) vs. 3 DOF (bottom).

Table 1: Decision Matrix: Leg DOF Configuration

Criteria	**Weight**	**2 DOF**	**3 DOF**
Cost	1	1	0
Mechanical Complexity	2	1	0
Ease of Control	3	1	0
Stability	4	0	1
Range of Motion	5	0	1
Weighted Total		6	9

We would select a 3 DOF system because, even in light of the increased complexity of a 3 DOF system, the benefits way outweighed the costs. The increased mobility and access to more different gaits allows us to better reach out end goals. With our high focus on safety, the additional DOF will allow us to also have a more stable position at "rest" or "stop" states as well.

4.2 Body Shape: Hexagonal vs. Rectangular Base

Figure 4: Design alternatives for body shape: rectangular (top) vs. hexagonal (bottom).

Table 2: Decision Matrix: Body Shape

Criteria	**Weight**	**Hexagonal**	**Rectangular**
Leg Mounting Symmetry	1	1	0
Manufacturing Ease	2	0	1
Rider Seating (2 people)	3	0	1
Transportability	4	0	1
Weighted Total		1	9

We would decide to go with a rectangular base in favor for both manufacturing and rider seating. This would allow for the most "space" per area for sitting and still allow each rider to be comfortable, while also still having a stable setup.

4.3 Microcontroller: Arduino vs. ESP32 vs. STM32

For the low-level motor control and communication, we evaluated three microcontroller platforms.

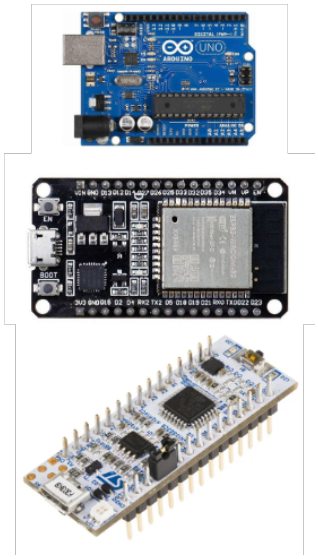

Figure 5: Microcontroller options considered: Arduino Uno (top), ESP32 (middle), STM32 (bottom).

Table 3: Decision Matrix: Microcontroller Selection

Criteria	**Weight**	**Arduino**	**ESP32**	**STM32**
Cost	1	1	1	0
Community Support	2	1	1	0
Processing Speed	3	-1	0	1
Wireless (BT)	4	0	1	0
PWM Channels	5	-1	1	0
Weighted Total		-5	12	3

In the end, we would decide upon using an ESP32 for our main controller. Due to having ESP32 boards in stock already, we would not need to spend more money. The processing capabilities, while not the best out of the three, is still more than enough. The built in Bluetooth + Wifi connectivity is also a major plus for our custom controller. We will also be pairing this with a Raspberry Pi to run OpenCV for AprilTag detection.

4.4 Controller: Custom vs. Premade (Xbox/PS)

Figure 6: Controller options: premade Xbox controller (top) vs. custom controller concept (bottom).

Table 4: Decision Matrix: Controller Design

Criteria	**Weight**	**Custom**	**Premade (Xbox)**
Ergonomics	1	1	1
Cost	2	1	0
Reliability	3	0	1
Customizability	4	1	0
Weighted Total		7	4

We would end up deciding on a custom controller utilizing an ESP32 to allow natural pairing with ESPNOW. This would also allow us to easily create our own future macros for positions using custom buttons, alongside easily add any other switches and features as we see fit without having to mod an entire foreign software and board. Furthermore, with us having 24/7 access to 3D printers with CMU's Robotics Club, we will be able to create any case at any moment for any changes.

5 Cyberphysical Architecture

The cyberphysical architecture, shown in Figure 7, maps the functional subsystems to specific hardware and software components, detailing the flow of power, information, and control signals throughout the system.

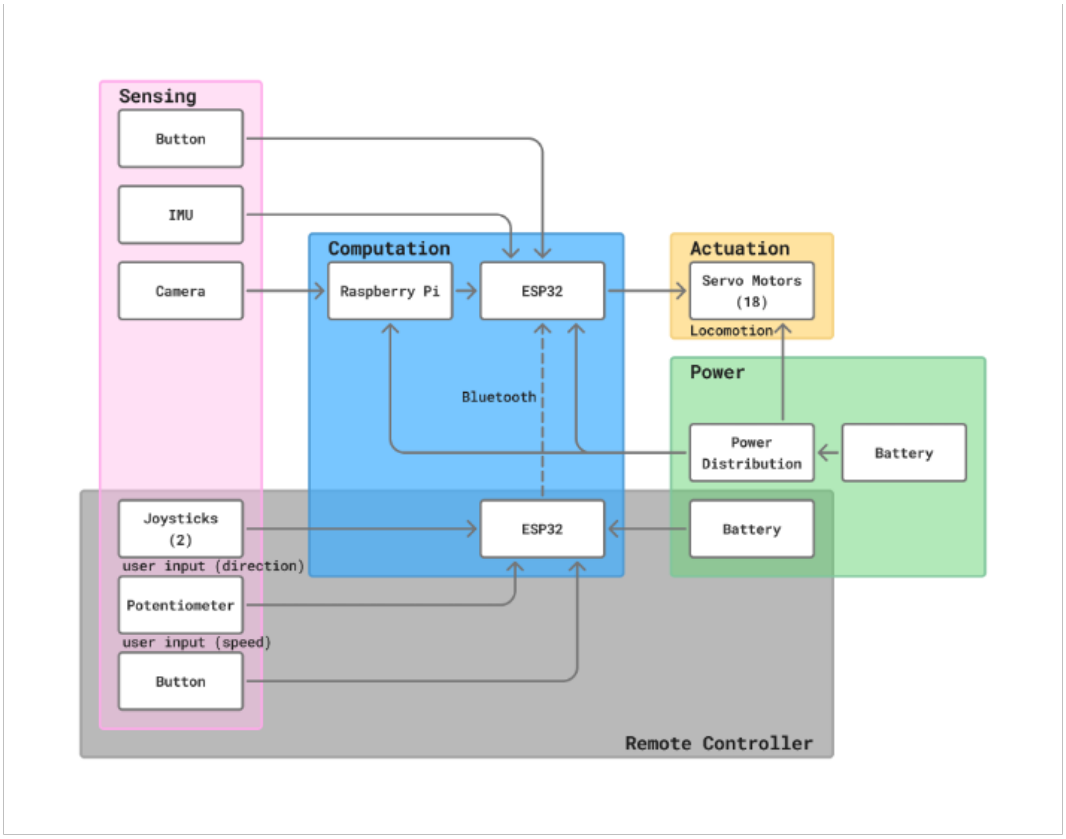

Figure 7: Cyberphysical Architecture Diagram. The system consists of Sensing (Button, IMU, Camera), Computation (Raspberry Pi + ESP32), Actuation (18 servo motors), Power (Battery + Power Distribution), and a Remote Controller (ESP32 + Joysticks + Potentiometer + Button) communicating via Bluetooth.

At a high level, the system consists of three main components: a sensing and input layer, a computational and control layer, and an actuation layer.

The sensing and input layer includes user input devices and onboard sensors. User input devices are placed on the remote controller and include: 2 joysticks for directional and rotational motion commands, a potentiometer for controlling the velocity, and a button to allow for selecting between autonomous and manual control.

Onboard sensors include an IMU to provide state estimation for stability monitoring and a downward-facing camera to be used for vision-based localization using AprilTags.

The computational and control layer serves as the core of the system and it includes vision processing and real-time control. A Raspberry Pi is used to process camera data and execute OpenCV-

based AprilTag detection algorithms. Real-time control is handled by an ESP32 mounted on the hexapod, which receives manual commands from the remote controller, processes onboard sensor data, and generates commands for the actuation system. A second ESP32 is integrated into the handheld remote controller to read user inputs and transmit control commands to the hexapod via Bluetooth.

The actuation layer consists of 18 servo motors that drive the hexapod's legs. Each leg is driven by 3 servos, providing 3 degrees of freedom per leg.

6 System Description/Depiction

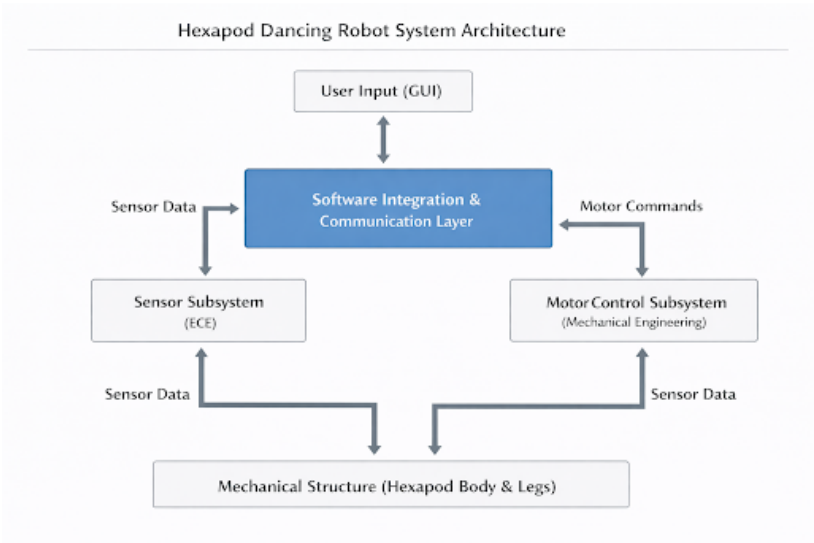

Figure 8: Hexapod Dancing Robot System Architecture. The Software Integration & Communication Layer serves as the central hub, coordinating between the Sensor Subsystem (ECE), Motor Control Subsystem (Mechanical Engineering), and the Mechanical Structure, with User Input (GUI) providing high-level commands.

In the above figure, Figure 8, provides an overview of the system architecture. The software integration and communication layer acts as the central coordinator, receiving sensor data from below (IMU, camera) and user commands from above (controller) or pre-coding commands, then dispatching motor commands to the actuation subsystem.

6.1 Motor Control Subsystem (Mechanical Engineering)

The motor control subsystem is responsible for generating the physical motion of the robot through servo motors. The hexapod uses 18 high-torque digital servo motors—specifically the Hitec D845WP—organized as three per leg:

- **Coxa joint (hip yaw):** Rotates the leg horizontally, enabling turning and lateral movement. One per leg (6 total).
- **Femur joint (hip pitch):** Lifts the leg up and down, critical for stepping over obstacles and generating the walking gait. One per leg (6 total).
- **Tibia joint (knee pitch):** Extends and retracts the lower leg segment, controlling ground contact and stride length. One per leg (6 total).

The D845WP servo was selected for its high torque output (694 oz-in / 50 kg·cm at 7.4V), coreless motor construction for smooth motion, IP67 water and dust resistance for durability, and steel gears with dual ball bearing support for long-term reliability under the high loads expected from a human-rideable platform. Each servo weighs 8 oz (227g) and provides 0.17 sec/60° speed at 7.4V, enabling responsive leg movements.

Each leg consists of three segments:

- **Coxa segment:** Mounted to the body frame, houses the coxa servo that provides yaw rotation.
- **Femur segment:** Connects the coxa to the tibia, driven by the femur servo for pitch rotation.
- **Tibia segment:** The longest segment, extending from the femur to the ground. A rod linkage connects the tibia servo to the lower leg, and a spherical foot at the tip provides stable, multi-directional ground contact.

The leg segments are fabricated from aluminum and 3D-printed high-strength polymer to balance structural integrity with weight. The combined leg reach provides adequate stride length and step height for smooth locomotion while supporting the required payload.

6.2 Sensor Subsystem (ECE)

The sensor subsystem provides information about the robot's operational state to monitor system behavior, support motion stability, and provide automation feedback:

- **IMU (Inertial Measurement Unit):** Mounted at the body center, provides 6-axis accelerometer and gyroscope data for orientation estimation and balance monitoring. Critically, the IMU enables flip detection—if the robot exceeds a dangerous tilt angle, the emergency shutdown is triggered automatically for safety.
- **Camera:** A camera connected to the Raspberry Pi provides a video feed for possible small-size computer vision model integration, such as waypoint navigation or obstacle detection.
- **Joysticks (on controller):** Two analog joysticks provide X/Y directional input for translation and rotation commands.
- **Potentiometer (on controller):** Provides continuous analog speed control, allowing the operator to smoothly adjust the robot's walking/dancing speed.

6.3 Software Integration & Communication Subsystem

The software subsystem integrates all hardware components and implements the robot's intelligence. Running primarily on the Raspberry Pi, it consists of several key modules:

- **Inverse Kinematics Engine:** Given a desired foot position (x, y, z) for each leg, the IK solver computes the required coxa, femur, and tibia joint angles using a geometric approach. This is the core algorithm that translates high-level movement commands into specific servo positions.
- **State Estimation:** Fuses IMU data (accelerometer and gyroscope) to estimate the robot's current orientation and detect unsafe conditions such as excessive tilt or flipping.
- **Gait Generation:** Implements multiple gait patterns:
  - Tripod gait: 3 legs move simultaneously for fast, stable walking.
  - Wave gait: one leg at a time for maximum stability at lower speed.
  - Custom dance gaits: choreographed leg sequences for performance.
- **Bluetooth Communication:** Bidirectional communication between the controller ESP32 and the onboard system, transmitting joystick/potentiometer values and mode commands at low latency.
- **Small-size CV Model (optional):** A lightweight Python-based computer vision pipeline on the Raspberry Pi that can process camera frames for target detection or navigation assistance.
- **Movement Design:** Predefined movement sequences and transitions that can be triggered via the controller, enabling smooth choreographed performances.

The software architecture follows a layered design: the Communication Layer handles Bluetooth I/O, the Control Layer processes user commands and sensor data, and the Execution Layer drives the servos through the IK engine and gait generator. This separation allows independent development and testing of each layer.

7 Project Management

7.1 Schedule

Figure 9 presents the project Gantt chart for the semester. Pre-spring break, team members work on their respective subsystems independently (CAD, firmware, sensors). Mechanical and electrical systems will be designed together to ensure proper wiring. Post-spring break, all subsystems are integrated and the full hexapod is assembled and tested.

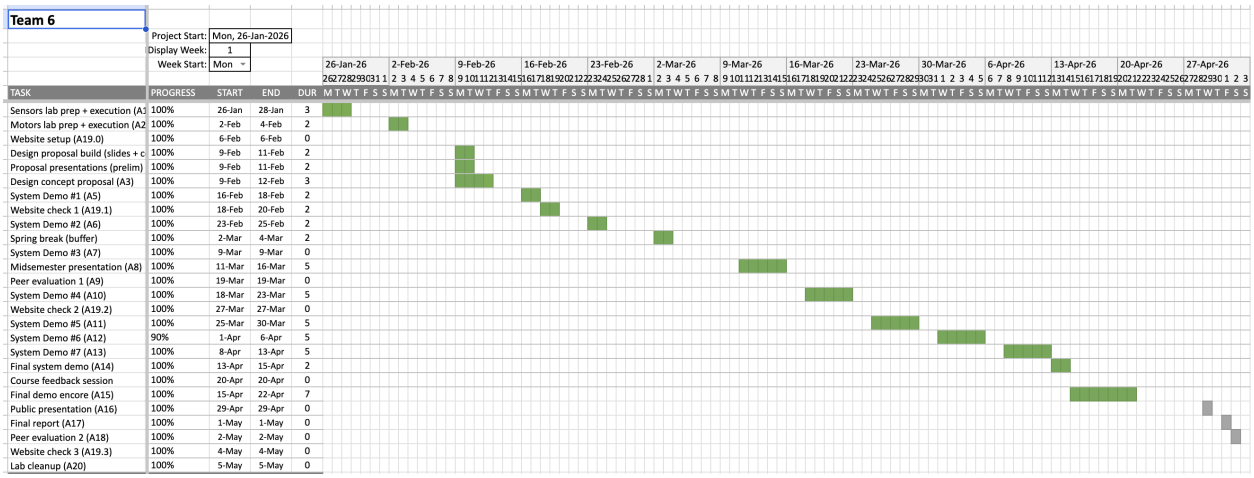

Figure 9: Project Gantt Chart showing the semester schedule from January to May 2026 with biweekly task granularity.

7.2 Team Member Responsibilities

Table 5: Team Member Responsibilities

Team Member	Primary Responsibility	Secondary Responsibilities
Thomas Wang	Software, System Integration	UI, possible CV/ML
Dillon Ma	CAD, Mechanical Design, Fabrication	Hardware, Wiring
Leena Aljurashi	Embedded Systems, Control	Integration, Electrical
Mithun Roy	Mechanical, Electrical	Power, Wiring

7.3 Budget

Table 6 provides a provisional parts list and budget. The servo motors represent the largest single expense.

Table 6: Provisional Budget

Subsystem	Item	Unit Cost	Qty	Total
Actuation	D845WP Servo (Hitec)	$134.99	18	$2,429.82
	PCA9685 PWM driver board	$8.00	3	$24.00
Compute	Raspberry Pi 4 (4GB)	$55.00	1	$55.00
	ESP32 DevKit	$10.00	2	$20.00
	Pi Camera Module v2	$25.00	1	$25.00
Sensing	IMU (MPU6050)	$5.00	1	$5.00
Power	Zeee 2S 7.4V 5200mAh 50C LiPo	$19.50	4	$78.00
	LiPo balance charger (iMAX B6)	$28.00	1	$28.00
	Power distribution bus bar	$5.00	1	$5.00
	LM2596 buck converter (5V)	$1.00	2	$2.00
	9V battery (controller)	$4.00	1	$4.00
Mechanical	6061-T6 Al square tube (1”×1”, 8ft)	$32.86	6	$197.16
	Nylon CF filament (1 kg spool)	$67.00	1	$67.00
	608-2RS bearings (10-pack)	$7.00	2	$14.00
	Fasteners (M3–M6 kit)	$15.00	1	$15.00
	Rubber foot tips, threadlocker	$12.00	1	$12.00
Controller	Joystick modules, pot., buttons	$20.00	1	$20.00
	ABS project enclosure	$9.00	1	$9.00
	Emergency stop button (22mm)	$7.00	1	$7.00
Misc.	Wiring, connectors, heat shrink	$40.00	1	$40.00
	Servo extension cables (20-pack)	$14.00	1	$14.00
			Parts Subtotal:	$3,095.98
			Sales Tax:	$216.72,
			Estimated Shipping:	$50.00
			Emergency Fund:	$137.30
			Total Estimated Cost:	**$3,500.00**

7.4 Risk Management

7.4.1 Design Risks

- **Servo torque insufficiency:** The hexapod is designed to carry two riders simultaneously, but the D845WP servo (694 oz-in at 7.4V) may only provide sufficient torque to support a single rider of up to 170 lbs. Supporting two riders could be hard.
  Mitigation: carefully calculate the weight limits.

7.4.2 Schedule Risks

- **Servo delivery delays:** The D845WP servos are specialty items that may have limited stock or long lead times. With 18 units needed, a supply shortage could stall the entire mechanical assembly.
  Mitigation: Place the servo order within the first week of the project. Identify at least one alternative servo with comparable torque specs as a backup. Begin software and IK development using a single servo for testing while awaiting the full order.
- **Spring break disruption:** The one-week break interrupts process
  Mitigation: Use spring break for asynchronous tasks or finished most part before spring break.

7.4.3 Resource Risks

- **Budget concentration in servos:** The 18 D845WP servos account for approximately $2,430. This leaves limited margin for other subsystems and emergency expenses.
  Mitigation: Finalize the servo selection early and lock in pricing. Evaluate other parts carefully and save some funds for emergency.

8 References

1. Hitec D845WP Servo Specifications, ServoCity. Available: https://www.servocity.com/d845wp-servo/
2. ServoCity Servo Gearboxes. Available: https://www.servocity.com/servo-gearboxes/